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# EECS 16A Touchscreen 1

**\*\*TA, UCS1, UCS1, UCS1\*\***

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# Semester Outline



Imaging  
Module



Touchscreen  
Module



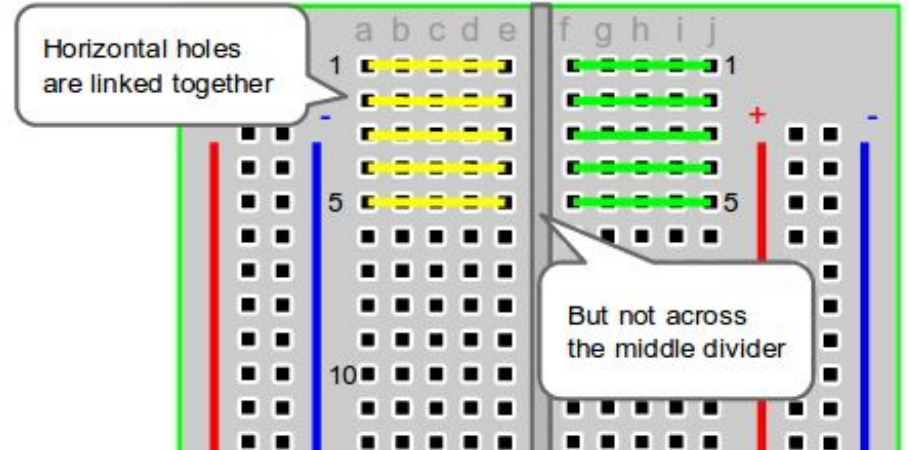
Acoustic  
Positioning  
Module

# Today's lab:

- Breadboarding
- New tool and circuit components:
  - Multimeter
  - Potentiometer
- Building multiple functional circuits

# Build up breadboarding skills

- Translate theoretical concepts from the lectures into practice:
  - **Resistors in series vs parallel**
  - **Ohm's law**
  - **Voltage Dividers**
  - **Kirchhoff's circuit laws**
- Very important skill set:
  - prototyping
  - debugging



# Poll time!

Review of breadboarding practices from Imaging 1.

1. Which of the following are good breadboarding practices?
  - a. Check the resistor value by its color bands
  - b. Plug in component legs in different rows
  - c. Use black wire for GND rail
  - d. Use red wire for GND rail
  
2. For which of the following components does polarity matter?

Resistor

LED

Capacitor

Ambient Light Sensor

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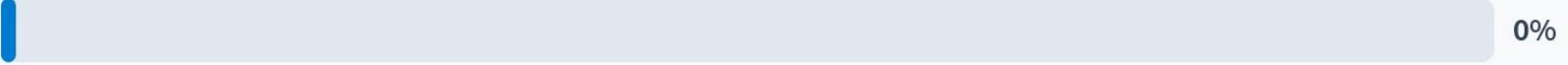
Ambient Light Sensor

## Which of the following are good breadboarding practices?

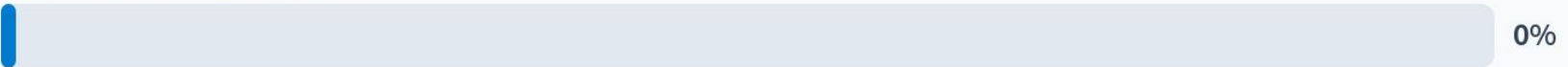
Check the resistor value by its color bands



Plug in component legs in different rows



Use black wire for GND rail

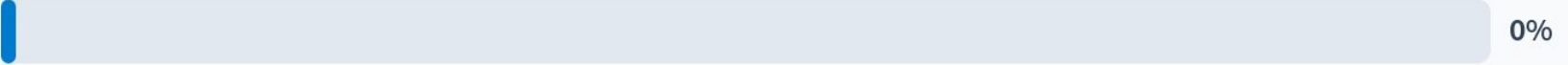


Use red wire for GND rail

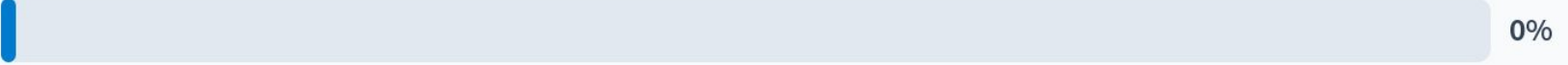


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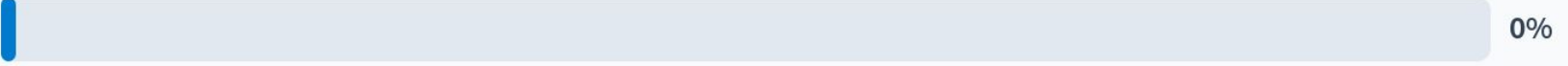
Resistor



LED



Capacitor

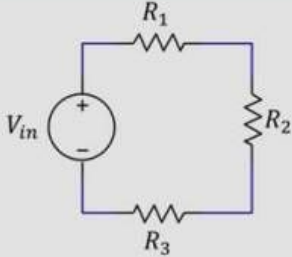
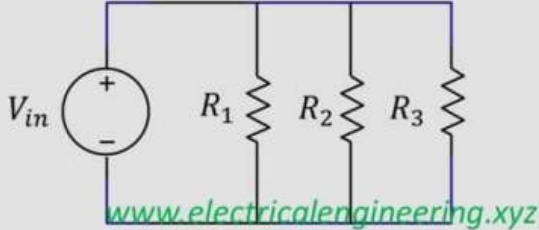


Ambient Light Sensor





# Theory of Resistor Configurations

|              | Series                                                                            | Parallel                                                                                 |
|--------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| How it looks |  |       |
| Voltage      | $V_{in} = V_1 + V_2 + V_3$                                                        | $V_{in} = V_1 = V_2 = V_3$                                                               |
| Current      | $I_{series} = I_1 = I_2 = I_3$                                                    | $I_{in} = I_1 + I_2 + I_3$                                                               |
| Resistance   | $R_{eq} = R_1 + R_2 + R_3$                                                        | $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$                       |
| Features     | If one components burns<br>current becomes inactive                               | If one component burns<br>current stops only through that<br>branch rest part works fine |

# Voltage Divider Circuit

What is the voltage value  $u_2$  at Node 2?

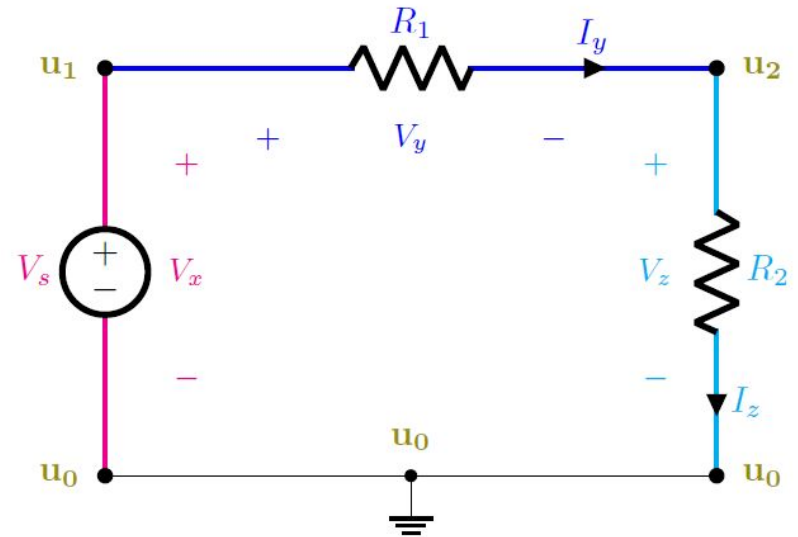
$$I_y = I_z = V_s / (R_1 + R_2) \text{ (Ohm's Law)}$$

$$u_2 - u_0 = R_2 * I_z$$

$$u_2 - 0 = R_2 * V_s / (R_1 + R_2)$$

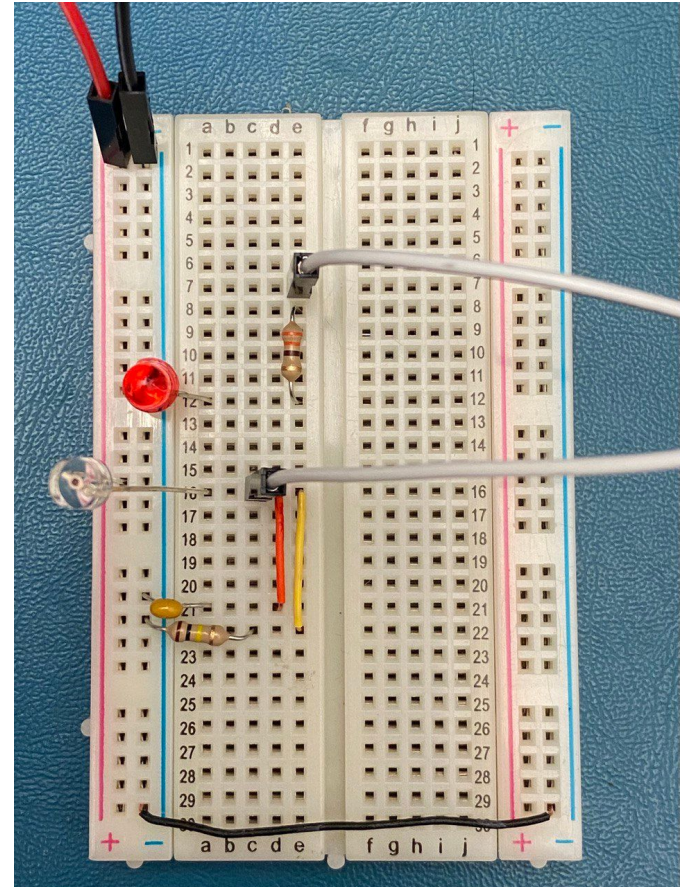
$$u_2 = V_s * R_2 / (R_1 + R_2)$$

What is the voltage value  $u_2$  if  $R_1$  equals to  $R_2$ ?



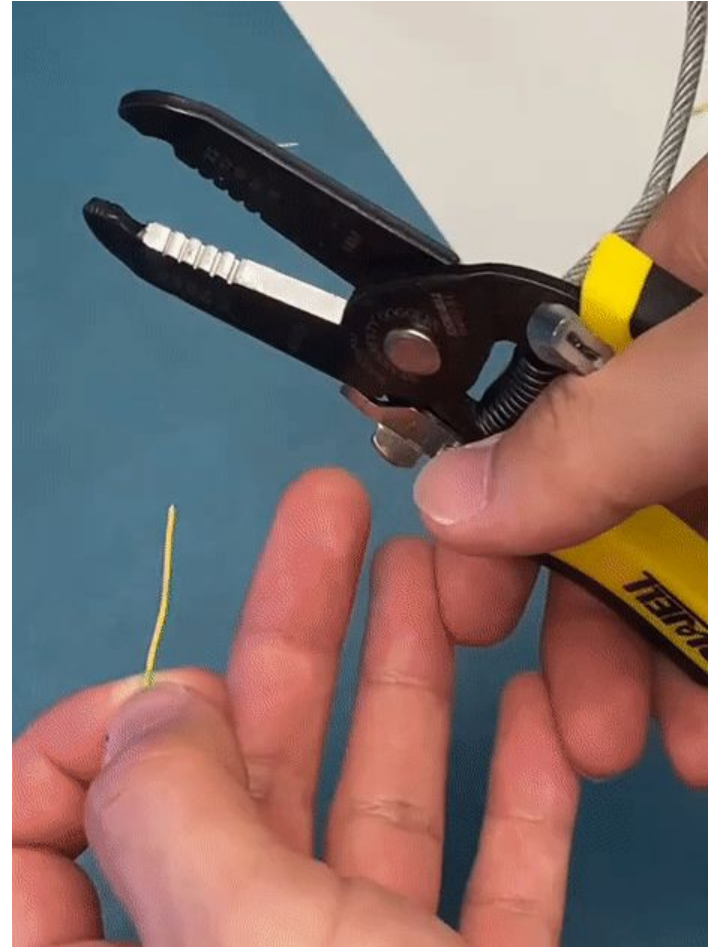
# Wiring components

- **Jumper wires** connect breadboard components to microcontrollers.
- **Standard wires** interconnect components on the breadboard:
  - wire box at the TA desk
- **Planar wiring** is important – we will refuse debugging your circuits otherwise.
- Interconnect GND rails to create common GND node.



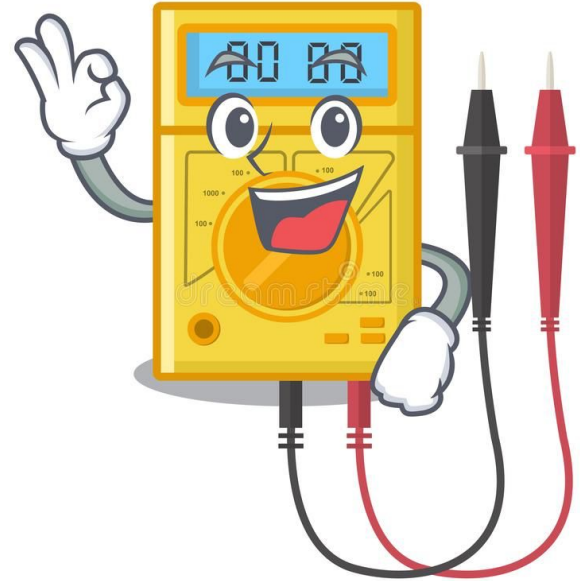
# Wire stripping

- Wire gauge (diameter):  
24 AWG = 0.5 mm



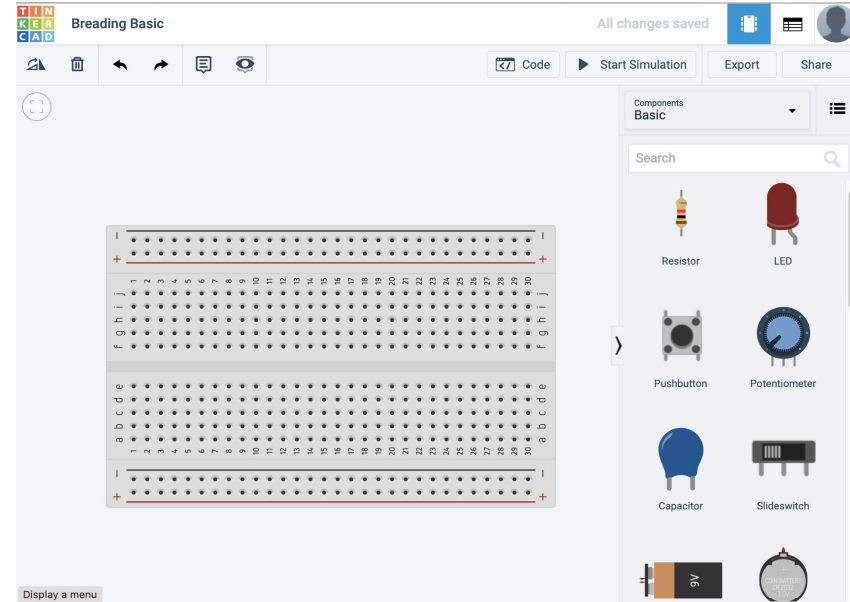
# Multimeter a.k.a. “Circuit Debugger”

- Voltmeter
  - Infinite resistance
  - Connect in parallel with component
- Ammeter
  - Very low resistance
  - Act as a wire in the circuit
  - Connect in series with component
- Ohmmeter
  - Remove resistor from circuit before use
  - Connect in parallel with resistor



# TinkerCAD

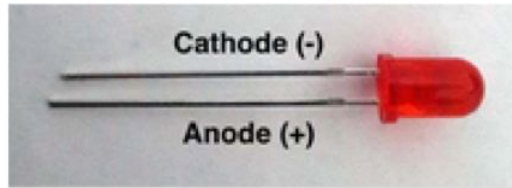
- Primary circuit simulator in 16A:
  - testing circuit behavior before implementing them
- Run online using an Autodesk account
- Multimeter is available





# Circuit Elements

LED



1.0 V

Red LED diode



1.4 V

Red LED diode

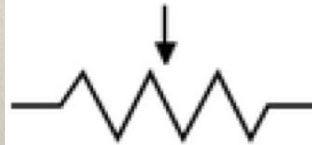


1.6 V

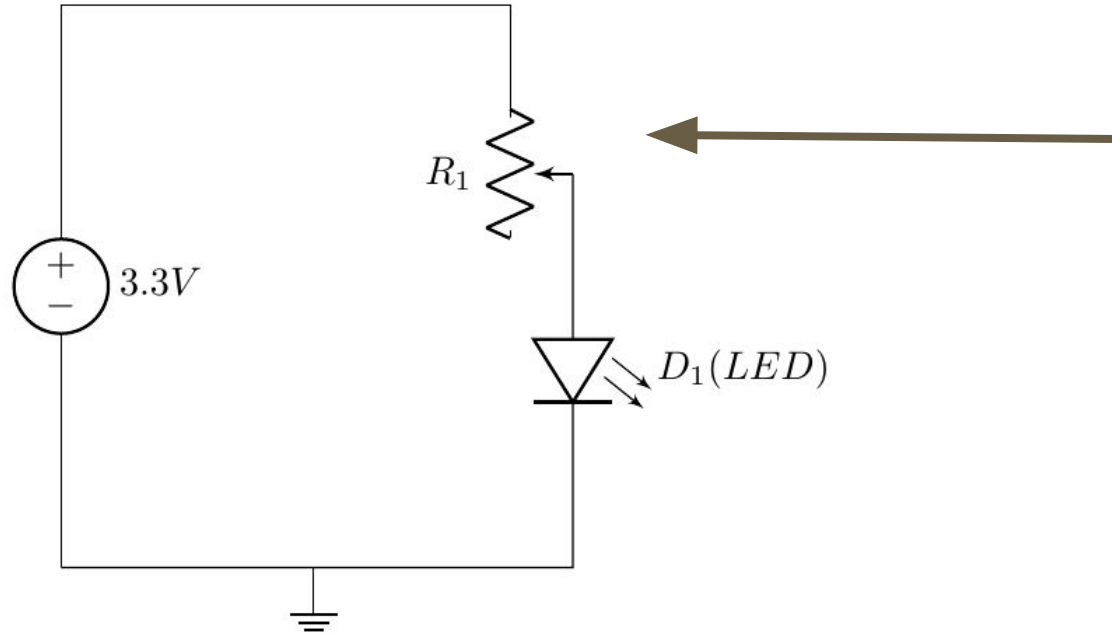
current



Potentiometer



# LED Fader Circuit





# Pointers

- Try to debug your circuit by yourself before you ask the TAs:
  - Ask for help if spending too much time
- 0.1 A, 3.3V limits
- Checkoff requirements:
  - Show how to use multimeter
  - Circuit is working
  - Correct wiring (planar, color convention, no jumpers)
  - Jupyter Notebook questions

Checkoff Form:



Feedback Form:



# Feedback

For lab training, submit your feedback here:

<https://tinyurl.com/fb-training-fa23>

Touch 1:

<https://tinyurl.com/fa23-touch1>

Reminder:

- TA Post-Lab Feedback:

<https://tinyurl.com/fb-ta-fa23>

